



Getting Started with Scrum

UCF COP 4934

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Sprint Events and Artifacts

Product Backlog – Product Goal

Sprint Planning

Sprint Backlog – Sprint Goal

Sprint with Daily Standups

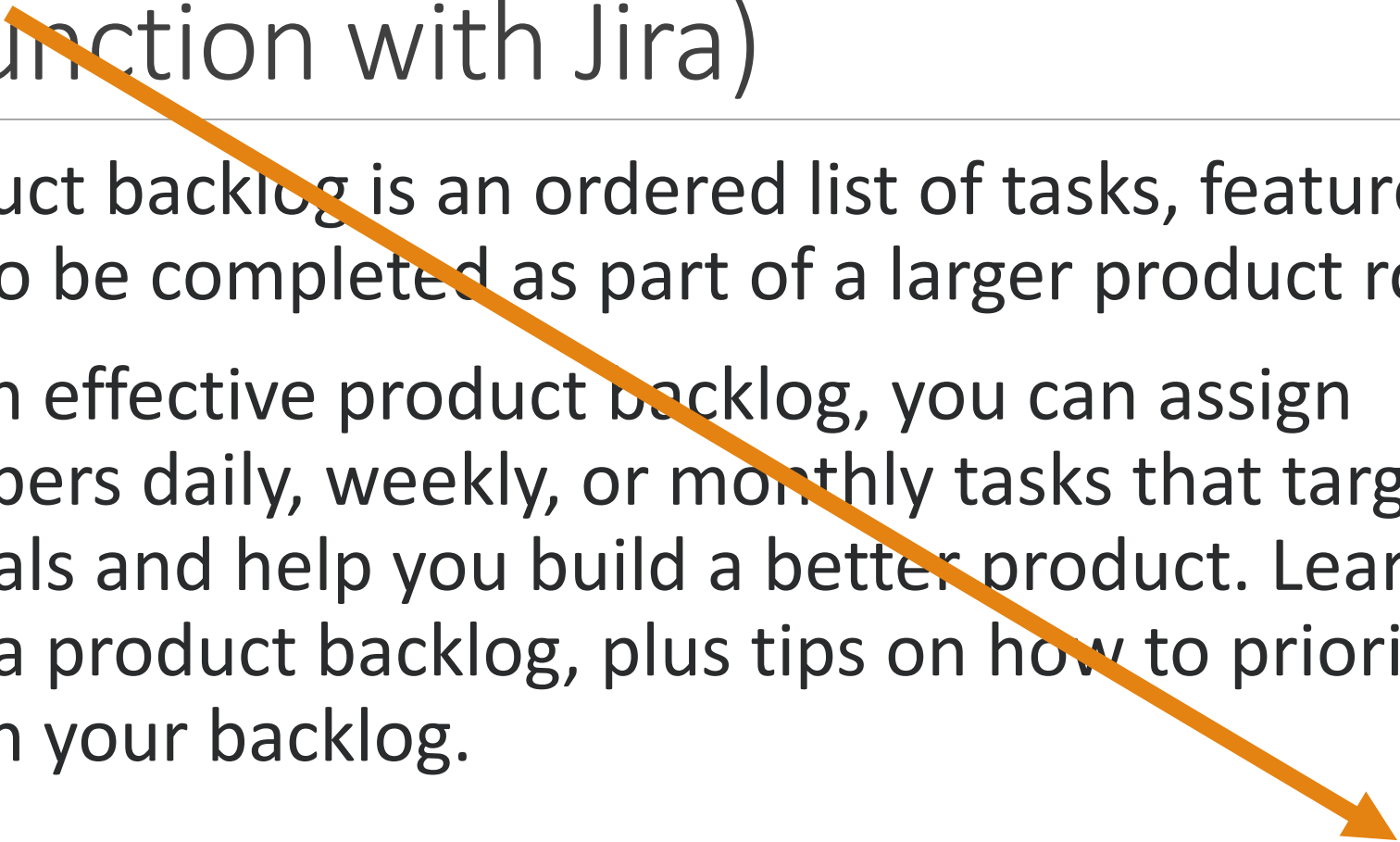
Sprint Review – Potential Increment

Sprint Retrospective

Step 1: Creating the Product Backlog (in conjunction with Jira)

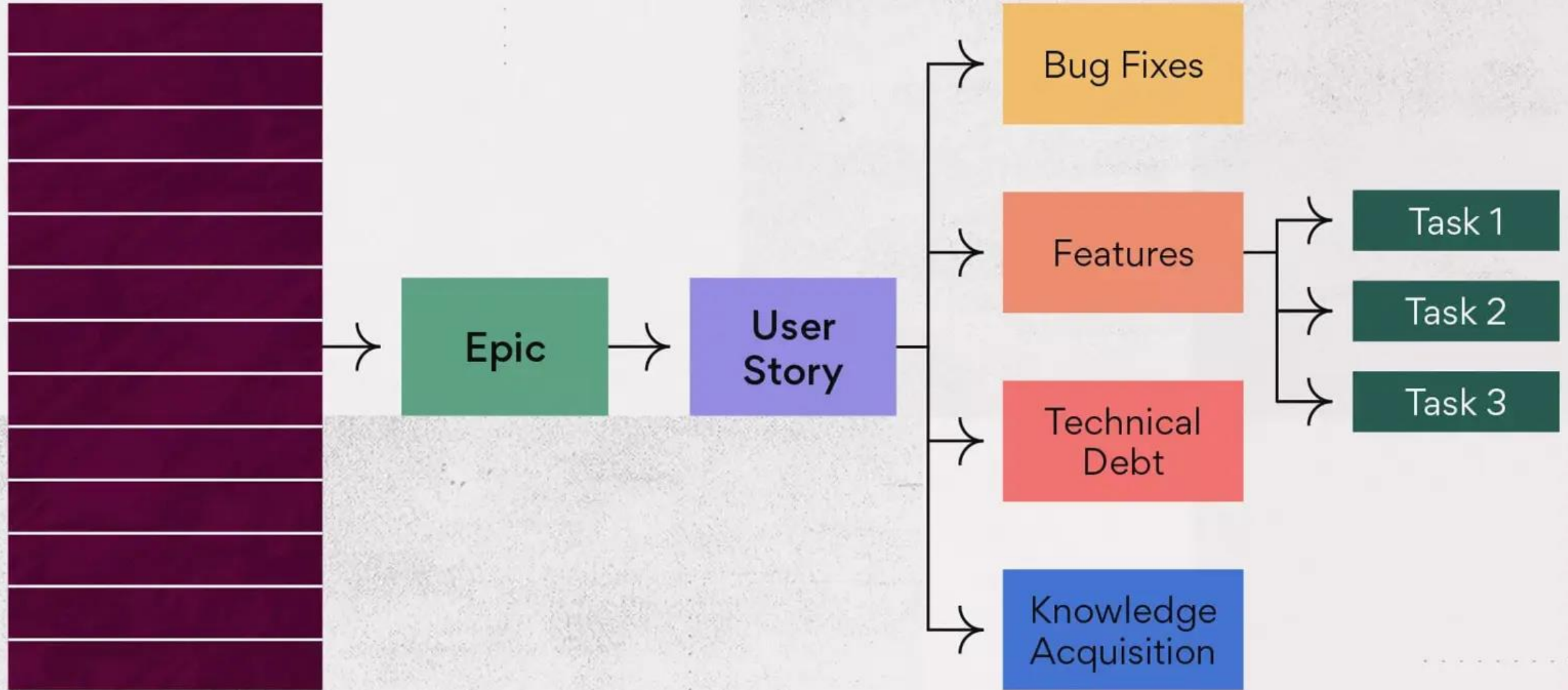
A product backlog is an ordered list of tasks, features, or items to be completed as part of a larger product roadmap.

With an effective product backlog, you can assign developers daily, weekly, or monthly tasks that target your end goals and help you build a better product. Learn how to create a product backlog, plus tips on how to prioritize the items in your backlog.



To Get Started

Product Backlog



Features

A feature, also known as a user story, is a function of the product that the product user finds valuable.

Features can be complex—often referred to as epics—or they can be simple.

Creating a story map can help your team determine what the user needs most.

Bug Fixes

Bug fixes are self-explanatory, and your Scrum team should address these quickly to uphold the integrity of the product. Some bugs may be important enough to interrupt your team's current sprint, while others can wait for the next sprint.

An overall rule with bugs, however, is to keep them at the top of your product backlog so your team doesn't forget about them.

Technical Debt

Technical debt, like financial debt, “accrues interest” when ignored.

When developers push technical work to the bottom of the product backlog, it builds up and becomes harder to accomplish.

Effective backlog management can prevent the buildup of technical debt.

When your team stays organized and takes on technical work in smaller, daily increments, you’re less likely to accrue interest on a huge piece of work.

Knowledge Acquisition

In knowledge acquisition, you gather information to accomplish future tasks.

Essentially, this is a research stage. When you identify a feature that needs more research, you create a knowledge acquisition task such as a prototype, experiment, or proof-of-concept to get the information you need to work on the feature.

Scenario: BrainBeats

Show scripted visuals in order to affect EEG-detectable frequencies/waves

Use EEG to detect brainwaves in various ranges.

Have computer acquire data via USB input.

Translate incoming data into music

- Include at least 6 methods of interpreting data to get music that can be selected.
- Each one is a dynamically loaded module
- Play sounds during music creation to form feedback loop

Save music to MIDI file.

Step 1 of 4: Build a Roadmap

The product roadmap is the foundation for the product backlog.

Your team should create a roadmap first, which will then serve as the action plan for how your product will change as it develops.

The roadmap is the vision for long-term product development, but it can also evolve.

Step 2 of 4: List Items

With your product roadmap in mind, your team can begin listing product backlog items.

These items should include both high-priority items and more abstract ideas.

During this phase of product backlog creation, you'll also need to communicate with stakeholders and listen to their ideas for product improvements.

If you're using the Agile method, you can organize this conversation as part of your sprint planning meeting.

Step 3 of 4: Prioritize

After your team lists all the product backlog items, sort and prioritize your most important tasks.

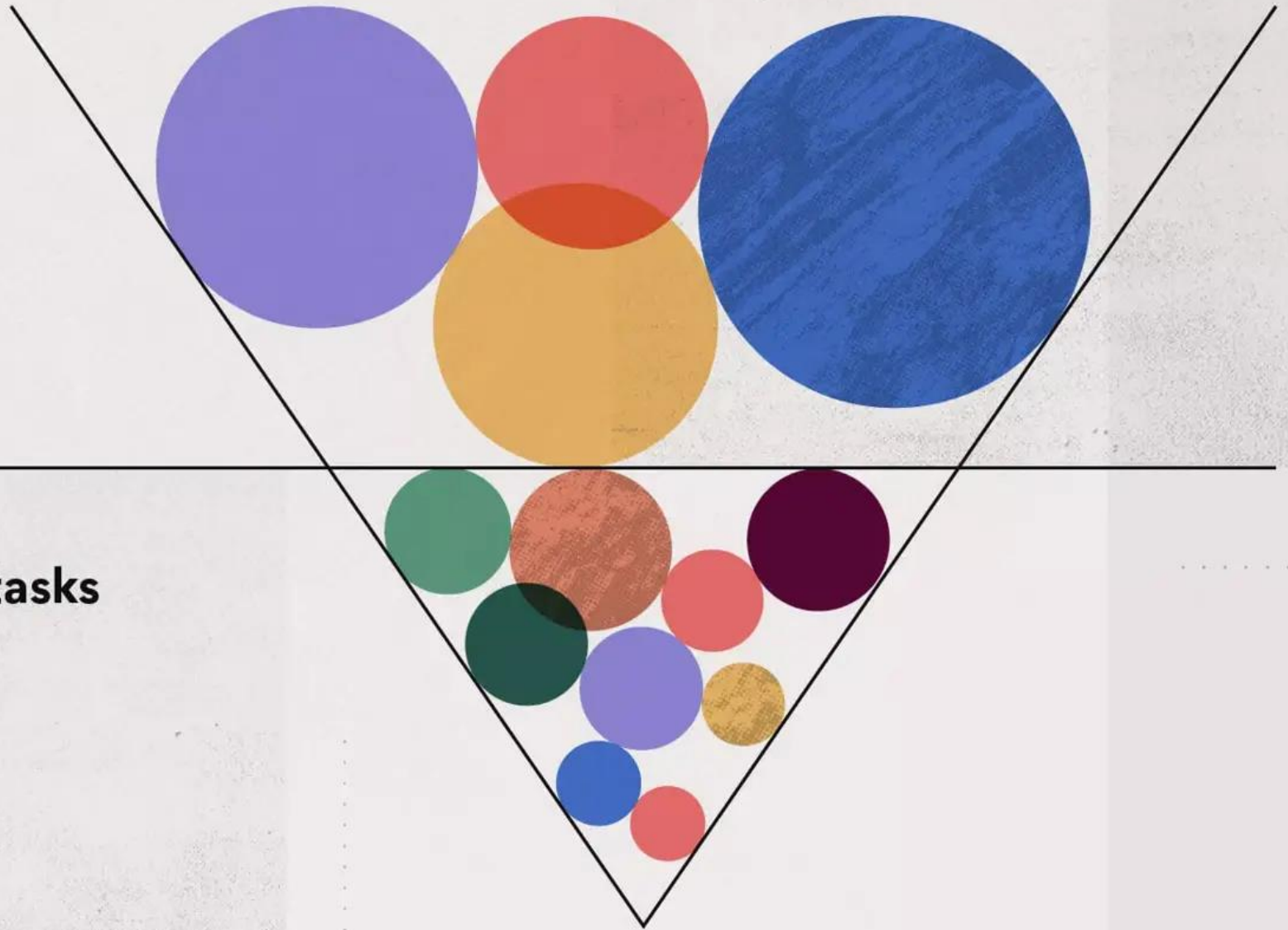
You can identify top-priority items by putting the customer front of mind and considering what items provide the most value to them.

Broad, vague ideas

Big ideas need time for refinement and re-prioritization.

Important, complex tasks

Prioritize sprint-sized items that are ready for planning.



Step 4 of 4: Update Regularly

As your team works through the product backlog, remember that it's a living document.

You can continuously add items to the backlog and prioritize or refine them as you work.

Consider This Example

Product backlogs look different between projects but some begin with an epic.

An epic is an overarching problem you're trying to solve for a customer.

First Epic

Epic: As a marketing manager, I want a content management system that allows me to deliver quality content to my readers.

This epic can help you work on user stories, such as how a user creates content in your new system or how they edit and share content with their teams. To continue our product backlog example, we can split the epic into more specific user stories.

Story 1: As a content creator, I want a content management system that lets me create content so I can inform customers about our products.

Story 2: As an editor, I want a content management system that lets me review content before it's published so I can ensure it's well-written and optimized for search.

Features

The product owner, Scrum master, and development team will determine features the product should include from the user stories and prioritize them based on importance.

Features the product should include for **Story 1**:

- Log in to the content management system
- Create content
- Edit a page of content
- Save changes
- Assign content to editor for review

Let Epics be your Guide

As the product manager, you'll use epics to guide your product roadmap and backlog list items.

As you can see with this example, one epic can result in multiple user stories and product features.

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Sprint Planning

Sprint planning is an event in scrum that kicks off the sprint.

The purpose of sprint planning is to define what can be delivered in the sprint and how that work will be achieved.

Sprint planning is done in collaboration with the whole scrum team.

What are We Planning

In scrum, the sprint is a set period of time where all the work is done.

However, before you can leap into action you have to set up the sprint.

You need to decide on how long the time box is going to be, the sprint goal, and where you're going to start.

The sprint planning session kicks off the sprint by setting the agenda and focus.

If done correctly, it also creates an environment where the team is motivated, challenged, and can be successful.

Bad sprint plans can derail the team by setting unrealistic expectations.

The What

The product owner describes the objective(or goal) of the sprint and what backlog items contribute to that goal.

The scrum team decides what can be done in the coming sprint and what they will do during the sprint to make that happen.

The How

The development team plans the work necessary to deliver the sprint goal.

Ultimately, the resulting sprint plan is a negotiation between the development team and product owner based on value and effort.

The Who

You cannot do sprint planning without the product owner or the development team.

The product owner defines the goal based on the value that they seek.

The development team needs to understand how they can or cannot deliver that goal.

If either is missing from this event it makes planning the sprint almost impossible.

The Inputs

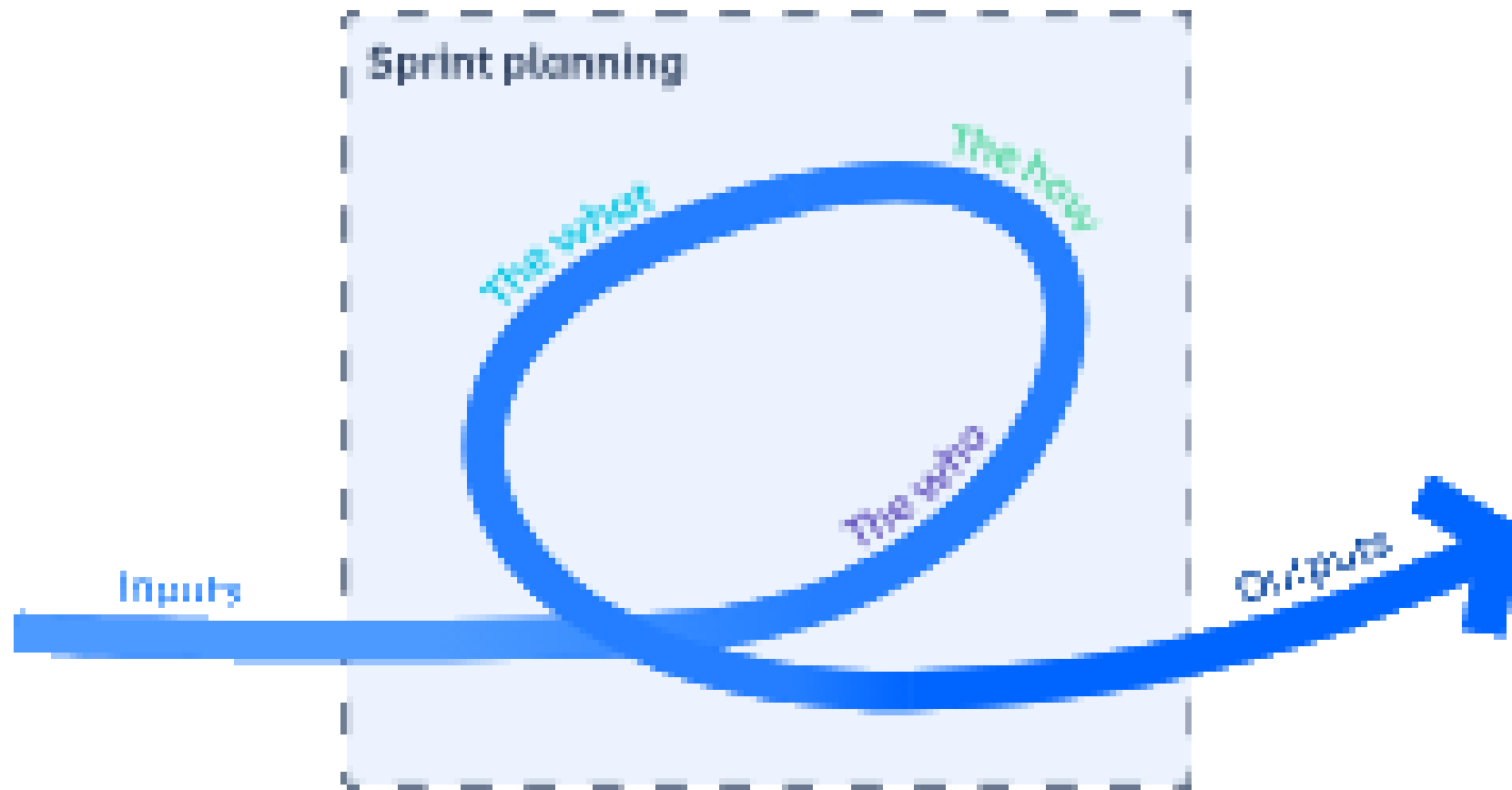
A great starting point for the sprint plan is the product backlog as it provides a list of 'stuff' that could potentially be part of the current sprint.

The team should also look at the existing work done in the increment and have a view to capacity.

The Output

The most important outcome for the sprint planning meeting is that the team can describe the goal of the sprint and how it will start working toward that goal.

This is made visible in the sprint backlog.



Prep for Sprint Planning Meeting

Running a great sprint planning event requires a bit of discipline.

The product owner must be prepared, combining the lessons from the previous sprint review, stakeholder feedback, and vision for the product, so they set the scene for the sprint.

For transparency, the product backlog should be up-to-date and refined to provide clarity.

Backlog refinement is an optional event in scrum, because some backlogs don't need it.

However, for most teams, it's better to get the team together to review and refine the backlog prior to sprint planning.

Time Box for Spring Planning

Sprint planning should be constrained no more than one hour for each week of the sprint.

So, for example, the sprint planning meeting for a two-week sprint would be no longer than two hours.

This is called "timeboxing", or setting a maximum amount of time for the team to accomplish a task, in this case, planning the sprint.

The scrum master is responsible for making sure the meeting happens the timebox is understood.

If the team is happy before the timebox is finished, then the event is over. A timebox is a maximum time allowed; there is no minimum time allowed.

Focus on Outcomes, not the Work

During sprint planning it is easy to get ‘bogged down’ in the work focusing on which task should come first, who should do it, and how long will it take.

For complex work, the level of information you know at the start can be low, and much of it is based on assumptions.

Scrum is an empirical process, meaning that you can’t plan upfront, but rather learn by doing, and then feed that information back into the process.

It's About the Goal

The sprint goal describes the objective of the sprint at a high level, but the backlog Items can also be written with an outcome in mind.

User stories are one great way of describing the work from a customer point of view. User stories, written like the one below, re-focus defects, issues, and improvements on the outcome the customer is seeking rather than the observed problem

Estimates – try but you might be wrong

Sprint planning requires some level of estimation.

The team needs to define what can or cannot be done in the sprint:
estimated effort vs capacity.

Estimation is often confused with commitments.

Estimates are by their very nature forecasts based on the knowledge at hand.

Techniques such as story points or t-shirt sizing add value to the process by giving the team a different way of looking at the problem.

They are not, however, magical tools that can find out the truth when there is none to be found.

The more unknowns, the less likely the estimate will be correct.

More About Estimates

Good estimation requires a trust-based environment where information is given freely, and assumptions are discussed in the pursuit of learning and improvement.

If estimates are used in a negative, confrontational way after the work is completed, then it's likely that future estimates will be either be much bigger to ensure they never are wrong again or the time taken to create them will be much longer as the team second guesses itself worrying about the implications of getting them wrong.

Scrum Planning Best Practices

It is easy to get so bogged down in the details of sprint planning you forget that the focus of sprint planning is to build a 'just enough' plan for the next sprint.

That plan shouldn't become a monkey for the team's back, instead, it should focus the team on valuable outcomes, and allow guardrails for self-organization.

A good sprint plan motivates everyone by defining an outcome and a clear plan for success.

But be careful planning too upfront.

Instead of building the most complete, "every minute of the sprint is accounted for" sprint plan, focus on the goal and build enough of a sprint backlog to get started.

Next, ensure that the product backlog is ordered to allow the team to pick up work if they delivered on the sprint goal early.

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The Sprint Goal

In the Scrum Guide, each Scrum artifact (Product Backlog, Sprint Backlog, Increment) has an associated commitment which provides information to enhance transparency and focus against which progress can be measured.

The commitments solidify the accountabilities of the Product Owner, Scrum Master and Developers.

The commitment of the Sprint Backlog is the Sprint Goal.

More About the Sprint Goal

The Sprint Goal is the single objective for the Sprint.

Although the Sprint Goal is a commitment by the Developers, it provides flexibility in terms of the exact work needed to achieve it.

The Sprint Goal also creates cohesiveness and focus, encouraging the Scrum Team to work together rather than on separate initiatives.

How Did We Get Here

The Sprint Goal is created during Sprint Planning by the Scrum Team and then added to the Sprint Backlog.

As the Developers work during the Sprint, they keep the Sprint Goal in mind. If the work evolves and is not what they expected (it is more or less complex than they thought, for example), they collaborate with the Product Owner to negotiate the scope of the Sprint Backlog within the Sprint without affecting the Sprint Goal.

The Sprint Goal does not change during a Sprint.

However, there are sometimes cases where the Scrum Team discovers something that invalidates the Sprint Goal.

When that happens the context of the situation or the environment the team is working in will influence what happens next, including possibly canceling the Sprint.

Sprint Goal Characteristics

When creating a Sprint Goal, it is important to keep in mind that it should be clear as to what the Scrum Team is aiming to achieve during the Sprint.

It is also important that the goal is attainable. As the Sprint Goal is in place to help create focus for the Scrum Team, the goal must be attainable and not too large or it will have the opposite impact.

More Characteristics

Remember, Scrum is about delivering value incrementally, so your Sprint Goal should be created with that in mind in order to maintain focus.

It is also important for the Sprint Goal to be visible so that the Scrum Team does not lose sight of it.

The purpose of the Daily Scrum is to inspect the progress toward the Sprint Goal, so it is important for Scrum Teams to maintain that practice.

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Daily Stand ups

Daily stand ups are the most popular format of meetings in the modern business world.

And for a good reason: they are easy to implement and offer instant benefits to anyone who runs them right.

But running standup properly is easier said than done: stand ups are easy to learn, but hard to master.

Daily standup is a short (up to 15 minutes) meeting with a defined set of questions to help team members exchange information about daily progress and blockers.

3 questions:

What did you do **yesterday**?

What will you do **today**?

Anything **blocking** your progress?

Participants:

up to **10 people**

Time limit:

up to **15 min**

Main Stand Up Principles

Keep daily standups short

Use the same set of questions every meeting

Conduct meetings daily and at the same time

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Sprint Review

Sprint reviews are a critical part of the Scrum process.

Without them, teams and clients cannot communicate efficiently, nor can they show everyone involved new iterations of working software.

Unfortunately, some teams bypass this step, confuse it with sprint retrospectives or do not maximize their meeting time.

What is a Sprint Review

A sprint review is an informal meeting where team members and the client discuss and show recently completed work.

It's not uncommon for there to be confusion around Scrum framework events.

Inexperienced teams can confuse an upcoming sprint planning meeting with a sprint review and a review with a retrospective.

Sprint Review Time Box

Like other Scrum events, a sprint review is time-boxed.

The review should last between 30 minutes and three hours, depending on how long the sprint lasted.

Since teams have limited time to discuss everything, the meeting facilitator, who is usually the

Scrum master, must ensure that all team members know what to discuss ahead of time.

More About the Sprint Review

During the meeting, the product owner explains the current state of the product backlog.

The product owner discusses what work has and hasn't been completed and updates delivery dates.

Developers should discuss what did and didn't work well and how problems were solved.

The product owner and development team can also show the latest working software iteration.

Continuing

Next, the team communicates and collaborates to determine if the market has changed enough to warrant revamping features, the final goal and what should be accomplished during the next sprint.

Budgets, timelines and project scope should also be discussed.

The scrum master should also remember to celebrate small wins and thank the team.

The Review Ends

By the end of the sprint review, the team should walk away with a cleaned-up product backlog that's ready for future sprints and a clearer idea of the work ahead.

Product Review Summary

Create an Agenda

Gather Data and Reports

Demonstrate the Product

Gather Feedback

Ask Questions

Refine the Backlog

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Sprint Retrospectives

Conducting a sprint retrospective allows you to learn from your previous efforts and improve future sprints, leading to streamlined Scrum sprints and a better end result.

Set the Tone

Setting the tone is the first step of conducting a sprint retrospective.

The goal of sprint planning is to collect feedback from different perspectives and use that feedback to set goals and improve Agile processes.

Everyone should feel like they have a voice at a sprint review, so set that tone from the beginning of the meeting.

You want to encourage people to share feedback and provide valuable insights during your sprint retrospective.

Focus on improving instead of blaming people for mistakes, and allow everyone to communicate openly and honestly.

Gather Feedback

Once you've set the tone and everyone knows what kind of input you're looking for, you can gather team members' feedback.

The 4 Ls retrospective is one way to gather feedback by asking what everyone has loved, loathed, learned, and longed for during a sprint.

If people have difficulty providing detailed feedback, using the 4 Ls provides a basic guideline.

There are various ways to gather feedback from everyone at your sprint retrospective.

You can have a group discussion if everyone is comfortable with that.

You can also have team members complete surveys or submit anonymous feedback that you can use to improve your next sprint.

Generate Insights

Collecting feedback is only part of the battle when conducting a sprint retrospective — you need to put that feedback into action.

Analyze the feedback you received to look for patterns and trends that may give you a better understanding of your team and the previous sprint.

These insights are a key part of setting goals, so encourage everyone to collaborate and contribute any valuable insights they may have.

We already mentioned how important it is to have different perspectives at a sprint review meeting, which becomes apparent when generating insights.



Create Action Items

Once you've learned more about your Agile team and your most recent sprint, you can turn those insights into actionable items.

This is where you can start setting goals and determine what changes to make for future sprints.

It's essential to set SMART goals — specific, measurable, achievable, relevant, and time-bound.

Setting SMART goals is crucial to achieving success and ensures that your goals are reachable and provide relevant results for your company.

Close Out the Retrospective

Now that you've taken steps to learn from your last sprint and improve your next one, you can close out your sprint retrospective meeting.

Provide a summary of key takeaways from the meeting, express your gratitude for everyone who contributed to the meeting, and give everyone a sense of closure.

When you leave your sprint retrospective meeting, you should have a clear implementation plan so your hard work doesn't go to waste.

Knowing how you will implement changes is a key part of sprint meetings.